Late Cretaceous to early Tertiary magnetostratigraphy of a continental sequence: Red Deer Valley, Alberta, Canada

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The Red Deer Valley exposes a conformable 350 m thick late Campanian to Early Paleocene fluvial–deltaic coal-bearing sequence named the Edmonton Group. Systematic sampling and paleomagnetic analysis of the dominant lithologies (mudstones, sandstones, and shales) were carried out over five sections correlated by coal seams to provide some 500 sites (stratigraphic levels). Primary detrital remanent magnetization directions after alternating field cleaning produce a polarity reversal pattern correlatable to sea-floor magnetic anomalies 29–33.

The Campanian–Maastrichtian boundary, based upon the first appearance of *Wodehouseia*, and the Maastrichtian–Paleocene (Cretaceous–Tertiary) boundary, based upon an Ir anomaly associated with dinosaur and Cretaceous palynofloral extinctions, correlate closely magnetostratigraphically with the marine foraminiferal section at Gubbio, Italy. Radiometric dating of bentonites has indicated the ages of these boundaries to be approximately 71 and 64 Ma, respectively.

Dans la vallée de Red Deer affleure une séquence fluviale–deltaïque contenant du charbon, d'une puissance de 350 m, désignée le groupe d'Edmonton, et d'âge compris entre le Campanien supérieur et le Paléocène inférieur. Un échantillonnage systématique et une analyse paléomagnétique des unités lithologiques dominantes (mudstones, grès et schistes argileux) ont été réalisés sur cinq coupes corrélées par des filons houillers fournissant quelques 500 sites (niveaux stratigraphiques). Les directions de l'aimantation rémanente détritique primaire, après traitement par champs alternatifs, présentent un motif de polarité inversée corrélatif avec les anomalies magnétiques 29–33 du plancher océanique.

La limite du Campanien–Maastrichtien, établie au moyen de la première apparition de *Wodehouseia*, et la limite du Maastrichtien–Paléocène (Crétacé–Tertiaire), fondée sur une anomalie Ir associée avec les extinctions des dinosaures et de la palynoflore du Crétacé, sont en étroite corrélation magnétostratigraphique avec la coupe riche en foraminifères marins à Gubbio, Italie. La datation radiométrique des bentonites donnent pour ces limites approximativement 71 et 64 Ma, respectivement.

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Introduction

The Red Deer Valley provides more than 100 km of continuous exposure of nearly horizontal strata ranging in age from late Campanian to Early Paleocene (Fig. 1). This succession, approximately 350 m thick, is fluvial–deltaic, becoming more fluviatile upwards. The principal lithologies, in order of decreasing abundance, are greyish olive floodplain mudstones, light grey fine- to medium-grained channel sandstones, backswamp carbonaceous shales and coals, grey to brown lacustrine shales, and siltstones. The sequence, named the Edmonton Group, is made up of the Horseshoe Canyon, Whitemud, Battle, and Scollard formations, from bottom to top (see Gibson 1977). This group is underlain conformably by the marine Bearpaw Shale and overlain conformably to disconformably by the fluvial Paskapoo Formation. There are no extraformational conglomerates in the sequence, no obvious deep channel scours, and no documented stratigraphic breaks of consequence.

The Red Deer Valley exposures are well known in vertebrate literature for latest Cretaceous dinosaur remains from the collections of B. Brown, the Sternbergs, and G. E. Lindblad (see Russell and Chamney 1967), and more recently for mammalian fossils (Lillegraven 1969). The top of the Maastrichtian (Cretaceous–Tertiary boundary) has been defined by the approximate level of dinosaur disappearance and by the level of marked Cretaceous palynofloral extinctions, notably the many species of the genus *Aquilapollenites* (Lerbekmo *et al.* 1979*a*,*b*; Srivastava 1970).

In addition to yielding dinosaur, mammalian, and microfloral remains, the Red Deer Valley sequence contains a considerable number of bentonitic volcanic ash beds suitable for radiometric dating (Lerbekmo *et al.* 1979*b*). Because of the combination of biological and radiometric data already available, this succession was chosen for paleomagnetic study in 1974 in an attempt to provide needed tie points between organic stratigraphy, geomagnetic polarity reversals, and absolute time. At the same time it was considered to be a test section to evaluate the practicality of paleomagnetic analysis as a routine stratigraphic tool in fluvial–deltaic sediments of this type. This paper is a follow-up to the preliminary results published on the upper part of the sequence, including the Cretaceous–Tertiary boundary (Lerbekmo *et al.* 1979*b*).

In order to sample the entire succession, five stratigraphic sections with horizontal separations of 10–27 km were used (Fig. 1) to provide about 500 sites (stratigraphic levels of sampling).

Methods

Sources of remanent magnetization

Edmonton and Paskapoo sandstones carry on the average about 0.5% opaque minerals, most of which are extractable with a hand magnet and therefore belong to the magnetite series (Chi 1966; Rahmani 1973; J. F. Lerbekmo, unpublished data). The finer lithologies are expected to carry a similar suite. The overall sandstone compositions indicate comparable quantities of volcanic, low-grade metamorphic, and plutonic or immature sedimentary material. Titanomagnetites and magnetites of mixed origin are expected to make up most of the highly magnetic suite.

The rocks have not been buried deeply (less than 1000 m), intruded, or folded. However, the mudstones are fractured at the surface on a scale of less than one to a few centimetres. The

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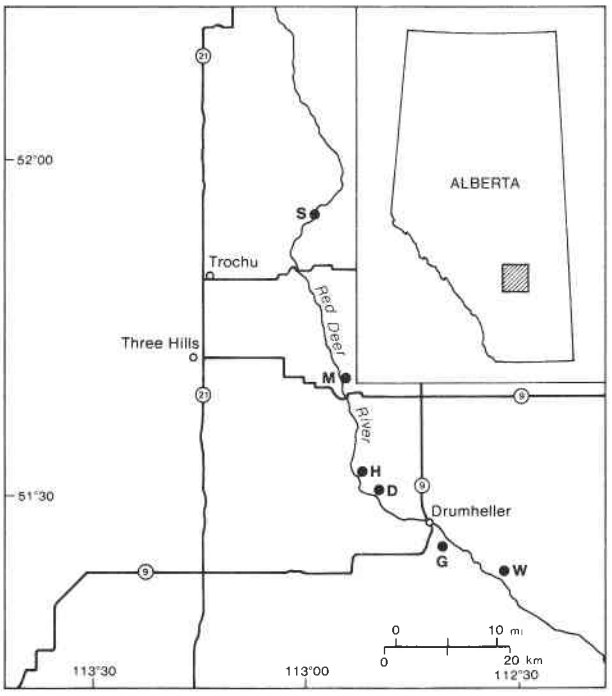


FIG. 1. Locations of sampled sections. S = Scollard Canyon; M = Morrin Bone-bed; H = Horsethief Canyon; D = Dinosaur Campground; G = *Gorgosaurus* Locality; W = Willow Creek. (After Gibson 1977, Fig. 1.)

fractures often have rusty-colored surfaces, no doubt coatings of iron hydroxides and possibly oxides. In addition, the sandstones occasionally have rusty-weathering concretions up to a few centimetres across of an iron-rich carbonate.

The valley walls are steep and eroding back rapidly. The valley was cut to a maximum depth of 150 m after the river was diverted southeastward by the last retreating Wisconsin ice lobe (Stalker 1960). Therefore the outcrop is not deeply weathered.

On the above information it was expected that the natural remanent magnetization (NRM) of these rocks would be made up basically of detrital remanent magnetization (DRM) caused mainly by the orientation of magnetite and titanomagnetite in the Earth's field at the time of deposition or soon afterward (see Verosub *et al.* 1979). Subsequently, with modest burial followed by re-exposure, such rocks may acquire secondary magnetizations: chemical remanent magnetization (CRM) from iron oxides and oxyhydroxides during diagenesis and weathering; and viscous remanent magnetization (VRM) from the effects of the recent normal field. Knowledge of details of the mineralogy is not directly applicable to a determination of the source of the NRM for, as Irving (1964, p. 4) has pointed out, the NRM is only a small fraction of the potential magnetization, and therefore the active component cannot be pin-pointed with certainty. The active component(s) is (are) usually better determined by the behavior of the sample during demagnetization, which will be treated later.

Sampling

The entire succession is highly bentonitic as a result of initial high proportions of glassy volcanic detritus, so only concretionary zones in the sandstones are suitable for diamond-drill coring. The mudstones and shales are fractured, probably because of incipient swelling near the erosional surface, and cannot be easily hand sampled. As all the lithologies are moderately soft, except where concretionary, a percussion tool was designed that was suitable for obtaining oriented samples. The tool will be described elsewhere. Field observations indicated that diagenetic concretions and weathering along fractures are very local, varying from sample to sample within a few centimetres at a single site. During early work five, and occasionally up to 10, independently oriented samples were taken at each site. Later the number was reduced to three for routine sampling. Sites were prepared by removing surface material until reaching rock that was clearly in place and undisturbed.

Early in the study it was decided to sample all lithologies, with the hope that a uniform sampling interval could be applied without lithologic restrictions. All lithologies proved to have a measurable magnetic direction after magnetic cleaning except for the coals, although the sandstones were more variable in behavior on the average than the mudstones and shales.

The first work was carried out on the upper part of the sequence (Scollard Formation) across the Cretaceous–Tertiary boundary (Lerbekmo *et al.* 1979*b*). Some intervals were sampled at vertical spacings of 10–20 cm in an effort to determine how much detail could be gleaned, particularly near polarity reversals. The choice of an optimum sampling interval to be applied to the rest of the succession was based upon the amount of magnetic detail that was likely to be useful in correlation. As a test of the reliability and usefulness of indicated short magnetostratigraphic polarity intervals, two sections of the reversed polarity zone (29r) in the Scollard Formation at Scollard Canyon were compared. The sections are on opposite sides of the river, about 3 km apart, and are closely correlated. The upper half of the reversed zone had been sampled at 10–20 cm intervals, and the lower half at 50 cm. These sections will be discussed in detail later. Suffice it to say here that thin, normal polarity intervals indicated in each section could not be correlated. Some of these intervals are indicated by only one site and may be the result of poor sample quality. Intervals supported by more than one site may be real but missing from the other section because of erosion or nondeposition.

We decided on the basis of this test area that polarity intervals less than 2 m thick could probably not be relied upon for correlation, and as a consequence routine site spacing was increased to 1.5 m for the remainder of the succession. Using 7.5 Ma for the length of the Maastrichtian, the average rate of sedimentation in this area was 1 m/30 000 – 35 000 years, and a 1.5 m site spacing thus represents roughly 50 000 years. Polarity zones documented by two sites therefore represent about 100 000 years and are probably useful for regional correlation.

Magnetic measurements

Alternating field (AF) demagnetization at 100–300 Oe (10–30 mT) is effective in removing VRM, but will not remove high-coercivity CRM, which would require thermal demagnetization. Because our sampling technique requires retention of the samples in plastic bottles throughout the measuring process, thermal demagnetization cannot be applied as a routine technique. Thus AF demagnetization was done in the expectation that it would be adequate for routine work. A two-axis tumbler similar to that described by McElhinny (1966) was used. Most of the early measurements were made on a Digico "complete results" spinner magnetometer; later ones were on an SCT three-axis cryogenic magnetometer.

The detrital magnetic component will vary in grain size and therefore magnetic quality (single domain versus multi-

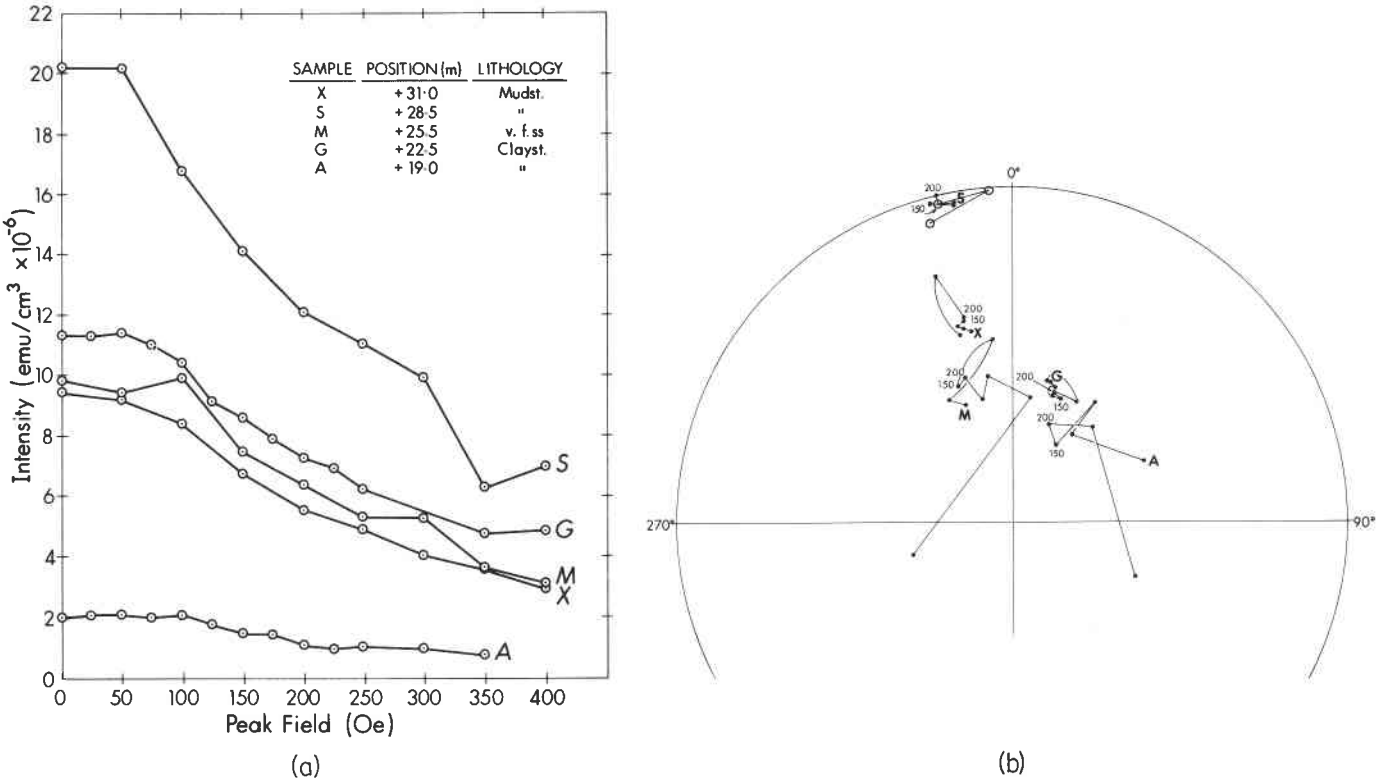
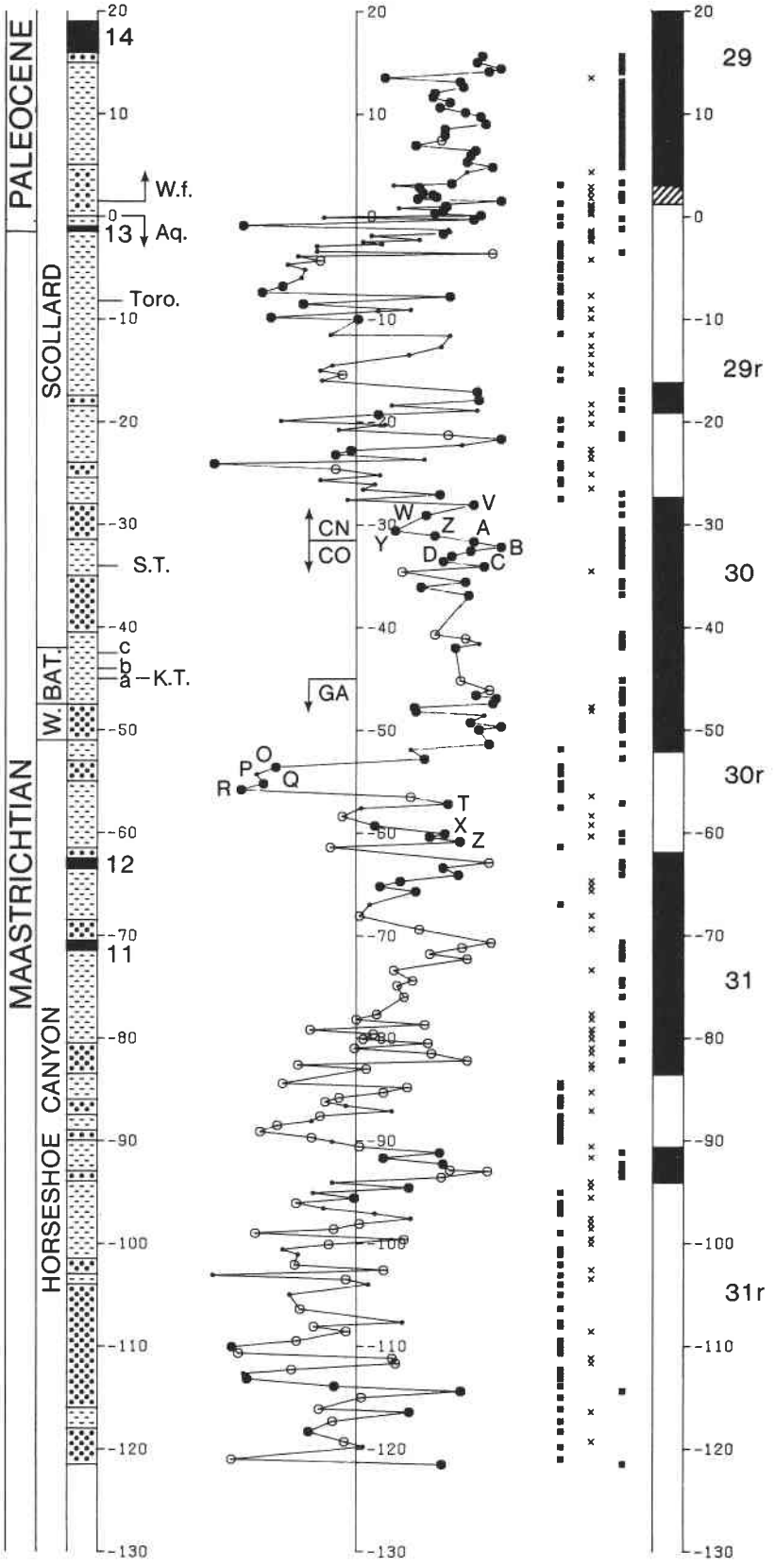


FIG. 2. Behavior during alternating field demagnetization of the first sample from each of five sites from the upper part of the Scollard Formation in the west Scollard Canyon section (see Fig. 3). (*a*) Decrease in intensity with increasing peak field. (*b*) Change in direction with increasing peak field treatment.

domain), as well as in concentration and probably composition, from mudstones and shales to sandstones. In addition, differences in permeability to fluids and in detrital major-component composition affect the degree and type of diagenesis and weathering in the different lithologies. For these reasons there is considerable variation in the strength of magnetization, and it is to be expected that different lithologies may behave differently during cleaning. On the average, the sandstones have the largest NRM, with a mean of about 2.6×10^{-6} emu/cm^3 (kA/m). However, they also have the largest proportion of soft magnetism, often largely removed by AF cleaning at 100 Oe (10 mT) peak field, reducing the remaining magnetic moment to about half of the NRM value (median destructive field). The mudstones and shales have a mean NRM of about 1.5×10^{-6} emu/cm^3 but have a smaller soft component, resulting in a more gradual decrease in magnetization during cleaning and a median destructive field between 100 and 200 Oe (10 and 20 mT). Fewer siltstones were measured, but they seem to have characteristics intermediate between those of the mudstones and sandstones, as might be expected. Most of the soft magnetism in these is lost at 100 Oe (10 mT), but very little at 50 Oe (5 mT). Coals are very weak magnetically and not suitable for analysis, as is also true for some highly carbonaceous shales.

The loss of moment with AF treatment for five pilot samples from five sites above the No. 14 coal seam in the west Scollard Canyon section is shown in Fig. 2*a*. The samples are from a normal polarity zone (29) (Fig. 3) and consist of two claystones, two mudstones, and a very fine sandstone. It will be noted that in this group the differences in magnetic moment are greater within the mudrocks than between mudrock and sandstone. However, strongly magnetized samples also contain large proportions of a low-coercivity component, which is lost most rapidly between 100 and 200 Oe (10 and 20 mT), at times at even lower treatments. Despite these differences in moment, however, all samples have relatively stable directions up to peak treatments of 250 Oe (25 mT) or higher (Fig. 2*b*). The stability index $d\theta/dH$ (Symons and Stupavsky 1974) is minimal between 150 and 200 Oe (15 and 20 mT), indicating maximum stability in this range.

Stepwise AF demagnetization was normally begun at 75 or 100 Oe (7.5 or 10 mT) and continued in steps of 25, 50, 75, or 100 Oe (2.5, 5, 7.5, or 10 mT), depending upon the strength of the sample, until stability was reached, or until the remaining magnetic moment approached the sensitivity limit of the instrument. For most samples this limit was reached near 300 Oe (30 mT), but for some it was at as little as 75 Oe (7.5 mT) and for others at as much as 400 Oe (40 mT) or more. The position of the 70 Ma Late Cretaceous pole was near 190°E and 72°N (Irving 1979), which would produce a declination and inclination in the Red Deer River area of about 330° and 73°. The mean declination and inclination of all samples collected from the Scollard Formation were calculated for AF peak treatments of 100, 150, 200, and 250 Oe (10, 15, 20, and 25 mT). The results were, respectively, 359°, 60° (k = 3.0), 310.5°, 80° (k = 4.2), 347.5°, 64° (k = 2.9), and 307°, 79° (k = 3.9). Thus the 200 Oe (20 mT) treatment provided the best agreement with the 70 Ma Late Cretaceous pole, but the scatter was large at each treatment level. It was decided that a peak field of 200 Oe (20 mT) was probably the optimum demagnetization treatment for the average sample and that the magnetic direction obtained at this treatment would be used for polarity determination, as described below. Though 150 Oe (15 mT) gave a slightly higher Fisher k, it was felt that there


PALEOCENE
MAASTRICHTIAN
SCOLLARD
W. BAT.
HORSESHOE CANYON
14
13
12
11
W.f.
Aq.
Toro.
S.T.
c
b
a
K.T.
CN
CO
GA
V
W
Z
A
Y
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29
29r
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SITE VGP LATITUDE
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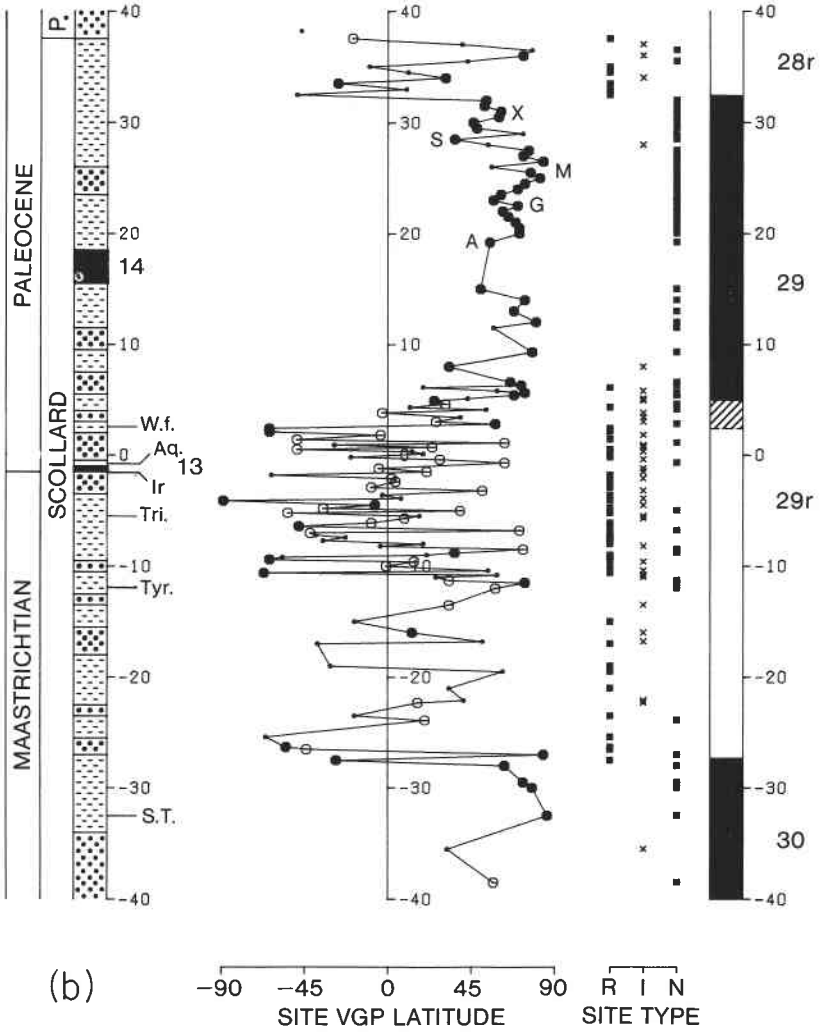


FIG. 3. Scollard Canyon sections. W. = Whitemud Formation; BAT. = Battle Formation; P. = Paskapoo Formation; K.T. = Kneehills Tuff; S.T. = Scollard tuff; W.f. = lowest occurrence of palynomorph *Wodehouseia fimbriata*; Aq. = highest occurrence of palynomorph *Aquilapollenites*; Ir = iridium anomaly; Tyr., Toro., and Tri. = highest local occurrences of *Tyrannosaurus*, *Torosaurus*, and *Triceratops*, respectively). Lithologic symbols: black = coal; dashes = mudrock; dots = sandstone. Site symbols: large dot = site with $N = 3$, which satisfies Watson's (1956) test for nonrandomness; open circle = sites with $N < 3$; small dot = site not satisfying Watson's test; R, I, N = reversed, indeterminate, and normal polarity sites. See text for fuller explanation. Right-hand column is interpreted correlation with numbered sea-floor magnetic anomalies (symbolism after Harland *et al.* 1982). Vertical scale in metres. (*a*) (*Opposite page*) East Scollard Canyon section (east side Red Deer River; centre, sec. 18, tp. 34, rge. 21, W 4th mer.). (*b*) West Scollard Canyon section (west side Red Deer River; NE 1/4, sec. 11, tp. 34, rge. 22, W 4th mer.).

was still the possibility of a contribution from a secondary component after this treatment. If demagnetization did not reach 200 Oe (20 mT), the highest treatment applied was accepted. In some cases this was as low as 75 Oe (7.5 mT).

Measurements were rejected if the measurement error (θ) was larger than 30°, though this restriction may have been somewhat more stringent than necessary. The value of θ may be determined from

$$\theta = \arctan\left(\sqrt{\Sigma(\delta Mi)^2}\,/\sqrt{\Sigma(Mi)^2}\right)$$

$$i = x, y, z$$

where δMi refers to the root-mean-square (RMS) error in the estimate of the ith component of M. A $d\theta/dH$ stability index screen is inappropriate for samples with unequal treatment steps, particularly when the steps are as small as 25 Oe (2.5 mT), because of the substantial errors in the measurement of weak samples. However, it was found that whenever the steps were 75 or 100 Oe (7.5 or 10 mT), the $d\theta/dH$ of these samples was nearly always less than 0.5°/Oe.

Polarity determination

One must accept that in the types of sediments being studied here even the initial DRM has a large scatter because of the limitations on the ability of the detrital magnetic grains to orient themselves in the Earth's magnetic field during and after deposition. In addition, subsequent effects such as diagenesis and weathering are apt to produce changes that may obscure the original direction of magnetization. In any case, it must be decided how much angular deviation from the position of the Late Cretaceous pole will be acceptable for the virtual geomagnetic pole (VGP) of a site to be indicative of either normal or reversed polarity. This angle from the Late Cretaceous pole we have called the "polar angle." VGP's falling outside an acceptable polar angle were considered to be of indeterminate (rather than intermediate) polarity. A polar angle of 70° divides the

Earth's surface into three approximately equal area zones, and as there are three polarity classes (normal, reversed, and indeterminate), a polar angle of 70° has some merit. However, each sequence of sediments has its own magnetic characteristics caused by variations in conditions of sedimentation, detrital composition, burial diagenesis, and weathering conditions. Therefore each probably has its own optimal "polar angle cutoff." One way of arriving at this optimum is to compare two sections close enough together to have undergone the same history and determine what scheme provides the best correlation between them. To this end we looked at the data from the two sections of the Scollard Formation referred to under "Sampling." Differences in sedimentation exist between the two sections, only 3 km apart, in the number and thickness of sandstones and sandstone/shale ratio (Fig. 3). Therefore real differences in polarity patterns may also exist, but they are probably minor because of the lack of obvious scour or intraformational conglomerates at the base of the sandstones and a general uniformity of fine grain size in the sandstones.

Comparing these two partial sections, the best correlation was achieved using a polar angle cutoff of 75°, but as there was little difference between 70° and 75°, we have adopted 70° as the polar angle cutoff in this paper. In a preliminary report (Lerbekmo *et al.* 1979*a*) an angle of 60° was arbitrarily picked, simply because it divided each hemisphere in half.

In addition, weathering during the past nearly 0.7 Ma has taken place under the present normal polarity field and will have produced some overprinting of normal polarity direction, for which compensation must be made. As in our earlier paper (Lerbekmo *et al.* 1979*a*), we have adopted a scheme similar to that used by Hillhouse *et al.* (1977) for Plio-Pleistocene fluvial sediments in Africa. If even one sample at a site records a reversed polarity, i.e., a polar angle within 70° of the Late Cretaceous reversed pole position, the site is judged to be reversed. The assignment of normal, reversed, or indeterminate polarity for a site was made objectively by computer programme. In the east Scollard section 73 sites were assigned reversed polarity by this scheme. Of these, 26 sites had all the samples reversed, another 25 sites had more than one reversed sample, or only reversed and indeterminate samples; 19 sites had one reversed, one normal, and one indeterminate sample; only three sites were assigned reversed polarity on the basis of one reversed sample and two normal samples; two of these three are in transition zones within 1 m of the selected polarity boundary position; the remaining site occurs in the upper part of zone 31r at an elevation of −97 m. The positions of the polarity zone boundaries have been chosen largely on the basis of the polarity assignments arrived at in this manner.

Discussion of results

Magnetostratigraphy

Introduction

Five sections were used to cover the interval from lowest Paleocene down to uppermost Campanian. The sections were selected from previous stratigraphic studies, especially those of Allan and Sanderson (1945) and Gibson (1977), who relied mostly on coal seams for correlation.

The VGP latitude shown for each site is the mean position obtained from the samples measured. This latitude is with respect to the 70 Ma Late Cretaceous paleopole position at 190°E and 72°N (Irving 1979). Site VGP latitudes are shown in three symbols, reflecting sample number and quality. This notation is similar to that used by Archibald *et al.* (1982) in Montana. In choosing polarity zone boundaries within what appear to be transition zones, most weight has been given to reversed polarity indications. Polarity changes not supported by more than one site have generally been disregarded, although some may be real. Indicated polarity intervals 2 m or less in thickness have also generally been omitted as being unreliable for correlation purposes, for reasons given earlier.

Scollard Canyon sections

The Cretaceous–Tertiary boundary occurs late in polarity chron 29r in the marine section at Gubbio, Italy (Roggenthen and Napoleone 1977; Lowrie and Alvarez 1977*b*) and in the continental Red Deer Valley section at Scollard Canyon (Lerbekmo *et al.* 1979*a*; Lerbekmo and Coulter 1984; and this paper). Therefore, the 29r–29 boundary is the most significant of the accepted time horizons for assessing the synchroneity of terminal Cretaceous extinctions. This boundary occurs in the two sections sampled at Scollard Canyon (Fig. 1), namely, west Scollard Canyon and east Scollard Canyon, on opposite sides of the river and about 3 km apart (Fig. 3 *a* and *b*). The western section displays the highest part of the sequence, whereas the eastern one gives the best continuous exposures of the lower part. Parts of both sections were sampled in detail (10–20 cm), especially near the Cretaceous–Tertiary boundary. Both sections show great variability in magnetic direction within reversed zone 29r but much more consistent directions in normals 29 and 30. Comparison of Fig. 3*a* and Fig. 3*b* suggests that the position of the 29r–29 boundary is somewhat different in the two sections. Because of the importance of the position of this boundary, we compared the two sections in detail in an attempt to choose the most reliable position. The interval from −10 to +15 m is shown at a larger scale in Fig. 4. The base of the 10 cm bentonite at the top of the Nevis (No. 13) coal is a precise time plane. However, the zero datum is 1 m above this bentonite in both sections, at the position chosen for the Cretaceous–Tertiary boundary in the east Scollard section, based upon the disappearance of *Aquilapollenites*. The lithology of the two sections over this 25 m interval is somewhat different: the west Scollard section contains more sandstone.

Sites SCE and F at west Scollard show good reversed directions and belong to zone 29r. SCT and sites above have consistent normal polarity and belong to zone 29. Sites in the intervening 2 m are classified variously as reversed, indeterminate, and normal sites under the scheme employed, but as a group have low VGP latitudes and might be best characterized as belonging to a transition zone of probable weak intensity.

Another approach to assessing polarity is to consider only those sites that show the least within-site variability, expressed by the α_{95} values. If only sites with $\alpha_{95} < 40°$ are considered, the transition zone in the west Scollard section contains only one site (SCQ). The magnetic directions of the highest 29r site (SCF), lowest 29 site (SCT), and the transition site (SCQ) are plotted on a stereogram in Fig. 5 along with the interpolated Cretaceous–Tertiary boundary (64 Ma) reversed and normal paleopole directions. Site SCQ gives a direction almost exactly midway between the SCF and SCT directions along a short great-circle path and probably locates the best mean position for the 29r–29 boundary at approximately +5 m.

Turning to the east Scollard Canyon section, it appears that the highest well reversed site is CMC, or perhaps CMA. CMA is the highest sample taken in a 60 cm black shale that overlies the Nevis coal bentonite and carries the highest occurrence of the Cretaceous palynomorph *Aquilapollenites*.

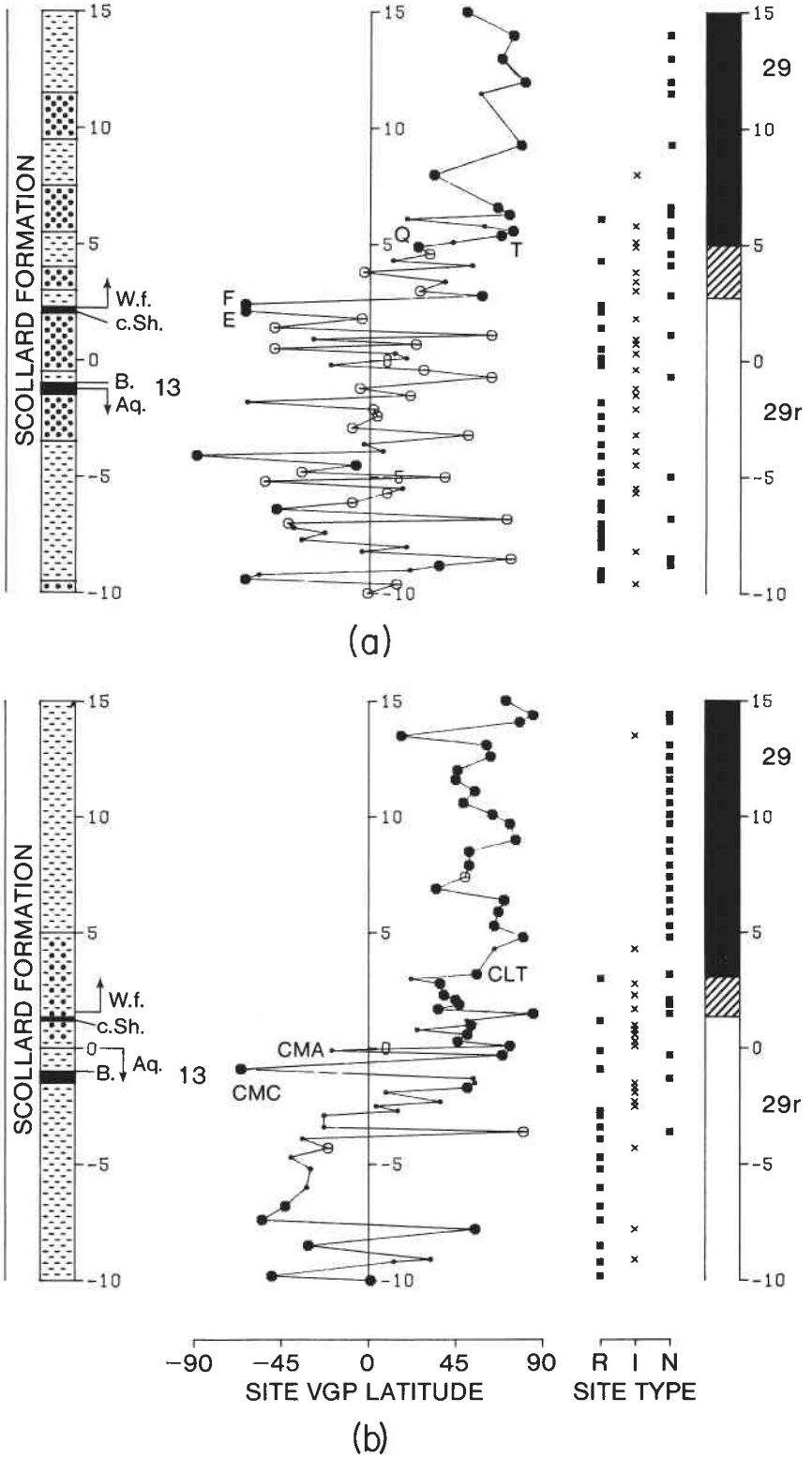


FIG. 4. Detail of the 29r−29 boundary interval in the (*a*) west Scollard Canyon and (*b*) east Scollard Canyon sections: c.Sh = coaly shale; B. = bentonite. See Fig. 3 for legend.

This shale is abruptly overlain by thinly interbedded sandstones and sandy mudstones. Comparison of the magnetic directions in the two Scollard sections suggests a hiatus in the east section relative to the west section of up to a few metres at the top of the black shale. However, there is a distinctive coaly shale 10−20 cm thick in both sections that is believed to be the same unit. It occurs 2 m above the bentonite at east Scollard and 3 m above at west Scollard, where almost the entire 3 m is sandstone. If the shale is indeed coeval, there is no more than 1 m of missing section in the east relative to the west section, and probably less, considering the likely differences between sedimentation rates of the sandstone and the mudstone. If there is less than 1 m difference in sedimentary record between the sections, some of the interval between CMA and CLT at east

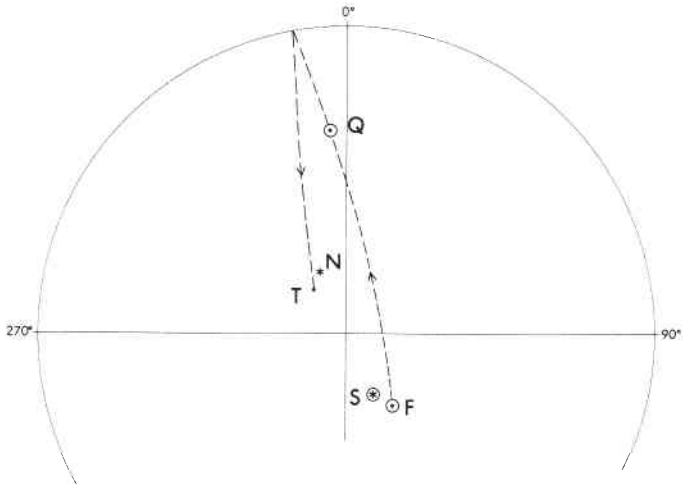


FIG. 5. West Scollard Canyon section 29r−29 boundary, showing directions of sites with $\alpha_{95} < 40°$ (SCF, Q, T), as well as directions of 64 Ma normal (N) and reversed (S) paleopoles (interpolated from 60 and 70 Ma paleopoles of Irving 1979).

Scollard must belong to reversed zone 29r, in which case it has been too stongly overprinted to show its primary reversed polarity. In any case, the west Scollard section is considered to more reliably display the polarity pattern near the Cretaceous−Tertiary boundary. Of course, there may also be a hiatus in the west Scollard section, but the fact that there is a dark shale of about the same thickness in both sections overlying the bentonite and underlying the first sandstone makes it doubtful that erosion has been significant at the base of the sandstones at this level. Evidence from a section at Ravenscrag Butte, Saskatchewan, also suggests this conclusion. There, apparently conformable mudstones and shales underlie and overlie the Ferris coal seam, the base of which approximates the Cretaceous−Tertiary boundary. The 29r−29 boundary occurs 3 m above the extinction level of Cretaceous palynomorphs (Lerbekmo 1985). Most of the 3 m is made up of the coal seam, however, which probably represents a slower depositional rate than the mudstones.

The lower boundary of 29r at Scollard Canyon is somewhat more distinct than the upper, occurring at about the same elevation in both sections (Fig. 3).

As mentioned earlier under "Sampling," comparison of 29r in the two Scollard sections does not permit correlation of what appear to be thin normal polarity zones in each section. The best documented of these is one near −12 m in the west and one near −18 in the east. Attention is drawn to the latter because it is at the same relative position as one found by Keating *et al.* (1975) at Deep Sea Drilling Project (DSDP) site 208 on the Lord Howe rise. These apparent normals may or may not be real, but failure to recognize them in adjacent sections led to the decision to increase sample spacing to 1.5 m for the rest of the Maastrichtian.

Polarity chron 30r is exceptional in that despite its short duration it is detected in nearly all sea-floor magnetic profiles, suggesting a strong dipolar field. The strongly reversed character of the upper part of 30r is well displayed (see sites O, P, Q, R) in the east Scollard section (Fig. 3*a*). The good quality of the samples in the upper part of 30 as well as in upper 30r and the presence of both sandstone and mudrock make this a useful interval for comparing the magnetic characteristics of a normal and a reversed zone (Figs. 6−9). In the normal zone both sandstones and mudrocks lose magnetic moment fairly similarly with increasing AF treatment (Fig. 6). The rate of loss is somewhat higher between 0 and 150 Oe (15 mT) than between 150 and 250 Oe (15 and 25 mT). The NRM moment of the sandstones is on the average somewhat higher and the within-site variation a little greater than in the mudrocks. The change in direction after AF treatment for the second sample at each of the sites in the normal zone is shown in Fig. 7. The effect is modest but somewhat larger and more variable in the sandstones than the mudrocks.

In the reversed zone (30r) there is a large difference in moment behavior between samples even though they are all very fine sandstones (Fig. 8). At two sites (GAO, GAR) a marked increase in moment takes place consistent with the removal of a secondary normally directed vector (VRM) from a reversed primary (DRM) direction. At the other two sites the increase in moment is very modest. Thus the moment contribution of the VRM to the NRM seems to be highly variable but the direction effect is strong in all samples (Fig. 9), most of which have highly inclined normal NRM's that move to moderately steep reversed inclinations after treatment at 150 Oe (15 mT).

The lower limit of 30r has been placed at about −62 m rather than somewhat higher for the following reasons. All three sites sampled in the west Scollard section between −55 and −63 m (−55.0, −57.3, −59.5) are strongly reversed (107°, −37°; 146°, −60°; 174°, −62°). Normal sites in this interval in the east Scollard section (GAT, X, and Z) at −57.2, −60.1, and −60.9 m all have directions near the present field (052°, 62°; 057°, 63°; 045°, 65°) and are therefore probably still overprinted. In addition, another section that contains this interval (Horseshoe Canyon) shows 30r to be at least 9 m and probably 16 m thick at that locality (Lerbekmo, in preparation). The sections both at east Scollard Canyon and Horseshoe Canyon show a mixture of directions in zone 31, suggesting that in this region it may be a mixed zone with some thin reversed intervals.

Although sample directions are rather highly scattered in the lower part of 31, the 31r−31 boundary is fairly well defined. A normal interval about 4 m thick seems to be present in the upper part of 31r, as evidenced by a number of sites. It has not been recognized as a separate sea-floor anomaly. It may be of limited geographic extent, as described for example by Harrison and Ramirez (1975), or perhaps has been included with anomaly 31 on the sea floor.

Tuff paleopoles—Tuffs present special paleomagnetic cases for determining "instant" paleopoles. There are at least five thin tuffs in the Scollard Canyon sections, the most prominent and widespread of which occur in the Battle Formation; they were named the Kneehills Tuff by Sanderson (Allan and Sanderson 1945). At Scollard Canyon the thickest tuff in the Battle Formation is about 20 cm thick. A second prominent tuff of similar thickness and characteristics occurs 13−15 m higher, in the Scollard Formation. It was informally called the "Scollard tuff" by Gibson (1977) (Fig. 3). Independently oriented hand blocks (4 and 3, respectively) were collected from these two tuffs, which are both in polarity zone 30 (Fig. 3). AF demagnetization showed the samples to be relatively stable between about 100 and 400 Oe (10 and 40 mT), with maximum stability near 200 Oe (20 mT). The magnetic directions of the tuffs at 200 Oe (20 mT) are shown in Fig. 10. The mean direction of the Scollard Tuff is 3.5°, 67.1° (k = 352), with α_{95} = 4.6°, whereas that of the Kneehills Tuff is 314.1°, 57.7° (k = 146),

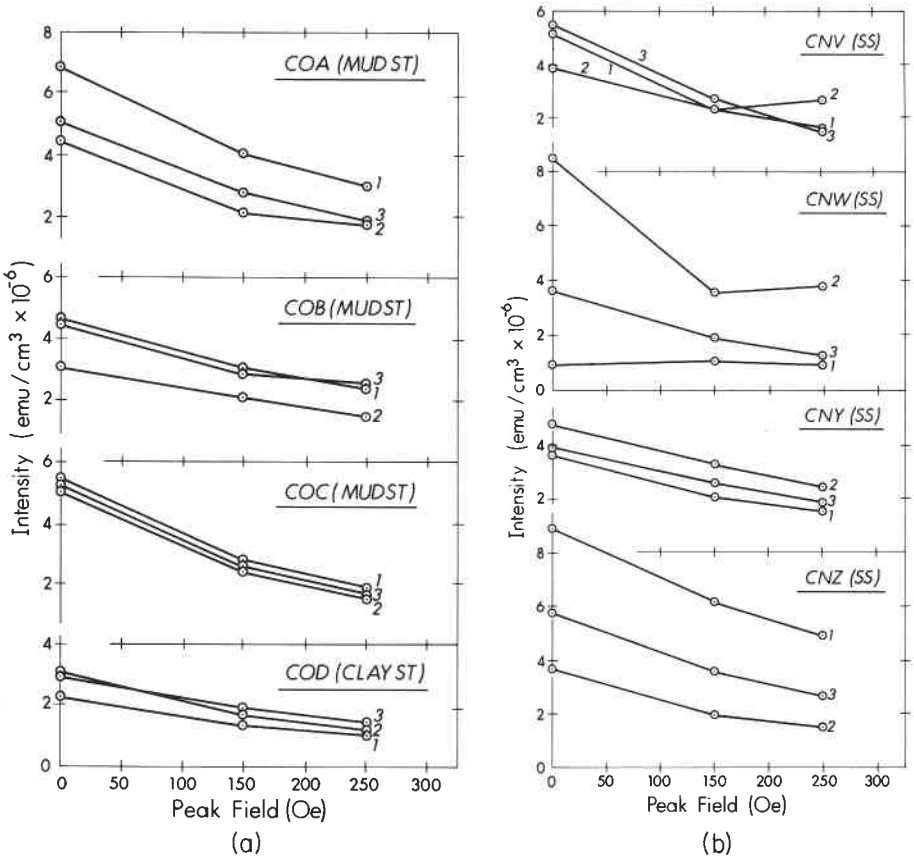


FIG. 6. Changes in intensity during alternating field demagnetization of the samples from eight successive sites from the upper part of polarity zone 30 in the east Scollard Canyon section. (*a*) Four mudrocks, COA, COB, COC, COD. See Fig. 5 for stratigraphic position. (*b*) Four sandstones, CNV, CNW, CNY, CNZ.

with $\alpha_{95} = 6.2°$. These paleofield directions represent instants in time and suggest a degree of secular variation comparable to that of today.

Morrin Bone-bed section

The tie between the east Scollard Canyon and Morrin Bone-bed sections (Figs. 3, 11) is not apparent from the figures. They are tied by the stratigraphic distance between the No. 10 and No. 11 coal seams. This separation was obtained from an intermediate section (Gibson's 1977 West Rowley section) where both seams are present.

Most sites in this section had only two of the three samples measured, because of uniformity of results from the first two. A tentative short normal identified as 32.1 (sea-floor anomaly 32A) would not be recognized in this section because it is supported by only one site. However, it was also indicated in a subsidiary section at Horsethief Canyon (Fig. 12). Fortuitously this normal is very close to the No. 10 coal. This seam was called the "marker seam" by Allan and Sanderson (1945) because they considered it to be the most stratigraphically reliable, though the thinnest, of their numbered seams. At Horsethief Canyon it occurs just above the brackish water fossils (*Corbicula* and *Ostrea*) of the "Drumheller marine tongue," which represents a last short advance of the Bearpaw Sea. The Morrin Bone-bed and Horsethief Canyon sections are 15 km apart, yet the position of the coal with respect to the normal polarity zone is identical, verifying Allan and Sanderson's interpretation of its stratigraphic value. The stratigraphic integrity of this seam relative to the others is likely the result of its shoreline-controlled environment. Other coals in this part of the sequence have linear northwest–southeast patterns related to floodplains of a number of streams of similar but not necessarily identical ages (Nurkowski 1982).

Dinosaur Campground section

Comparison of the overlapping parts of the Dinosaur Campground (Fig. 13) and Morrin Bone-bed sections indicates that the lowest reversed zone above normal 33 is partly missing in the Dinosaur Campground section. It is also noteworthy that the sites in zone 33 have rather uniformly high VGP latitudes, particularly below −220 m.

Gorgosaurus and Willow Creek sections

The consistent normality of the lower part of the Dinosaur Campground section continues downward through the shorter Gorgosaurus (Fig. 14) and Willow Creek (Fig. 15) sections. Detailed sampling was halted at the No. 1 coal seam in the upper Campanian. The 40 m interval underlying the No. 1 coal to the top of the marine Bearpaw Shale (Fig. 15) is almost entirely made up of channel sandstones of fluvial or shoreline origin. Reconnaissance sampling in the upper Bearpaw Shale at Willow Creek, East Coulee, and Dorothy (10 and 25 km farther southeast, respectively) indicates that polarity zone 33 continues downward to about −380 m. These results have been utilized in the construction of the summary polarity column of Fig. 17.

Cretaceous–Tertiary boundary paleopole

The lowest 10 m in polarity zone 29 in the east Scollard Canyon section has 22 sites, 20 of which have α_{95} values of less than 40°s. The 10 m represents about 300 000 years, based

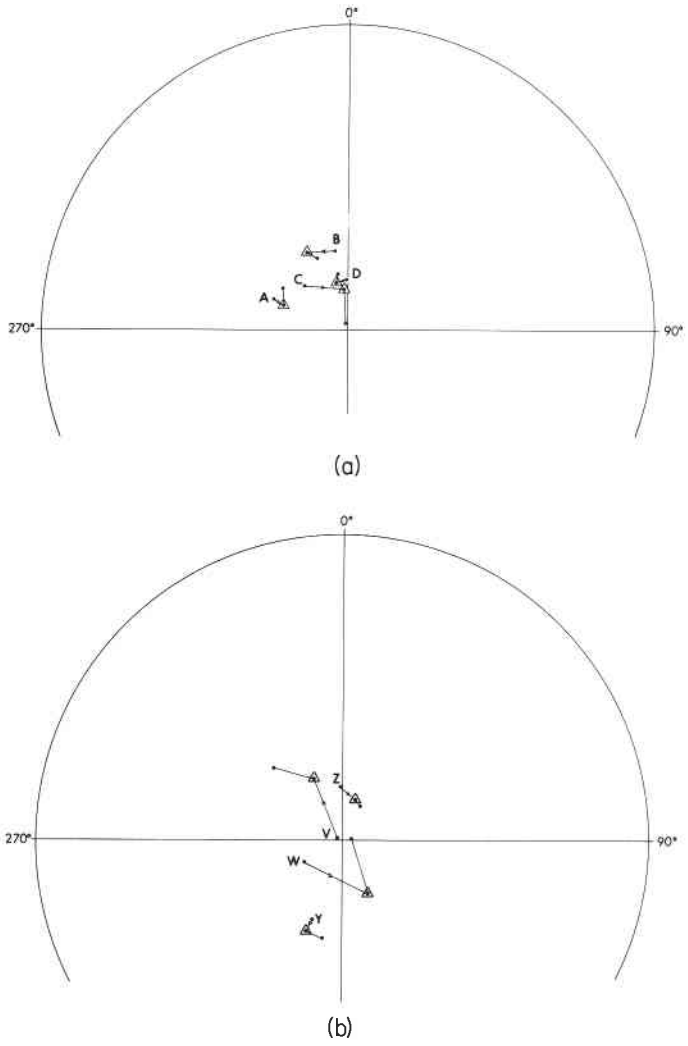


FIG. 7. Changes in direction during alternating field demagnetization of the second sample at each of eight successive sites from the upper part of polarity zone 30 in the east Scollard Canyon section. (*a*) Four mudrocks, COA, COB, COC, COD. Triangle marks direction accepted (after 150 Oe (15 mT) peak treatment). (*b*) Four sandstones, CNV, CNW, CNX, CNY. See Fig. 3 for stratigraphic position.

upon average rates of deposition of this section, a suitable time interval for determining a Cretaceous–Tertiary boundary paleopole. The mean direction for each of these 20 sites is shown in Fig. 16. The mean direction of the site means is 330°, 72° with $\alpha_{95} = 6.6$. The paleopole position calculated for this direction is 183°E, 72°N. The rocks being measured, just above the Cretaceous–Tertiary boundary, have an age of 63–64 Ma (Lerbekmo *et al.* 1979*a*; Baadsgaard and Lerbekmo 1980). Pertinent paleopoles given by Irving (1979) are 190°E, 72°N for 70 Ma and 187°E, 77°N for 60 Ma. The magnetic directions at Scollard Canyon that would produce these paleopoles are 330°, 73° and 339°, 72°. The interpolated paleopole for 63–64 Ma would be 188°E, 75°N, requiring a direction of 336°, 72° at Scollard Canyon. Both the 70 and 60 Ma pole directions are well within the α_{95} circle of confidence of the mean direction obtained at Scollard Canyon.

Summary section

Figure 17 is a composite section at reduced scale showing the polarity chron nomenclature of Harland *et al.* (1982) as well as the traditional sea-floor anomaly designations. Figure 18, for comparison, shows the magnetostratigraphy for this part of the geologic column at Gubbio, Italy, as well as the sea-floor anomalies from three ocean basins. It will be noted that the differences in relative thickness of the polarity zones between the Red Deer Valley and Gubbio are less than between the individual ocean-basin magnetic profiles, even though the comparison is between a fluvial–deltaic sequence and a deep-sea foraminiferal ooze.

Figure 19 shows the polarity time scale for this and adjacent parts of the column according to Harland *et al.* (1982), in which the Maastrichtian magnetostratigraphic and biostratigraphic data are based largely on the Gubbio section. Although the polarity pattern is unmistakably similar, there are important differences in the position and radiometric age of the Campanian–Maastrichtian boundary, to be discussed in the next section.

In the Red Deer Valley the consistency in magnetic direction of the samples in polarity zones 33 and 29 as compared with the interval between does not appear to bear any relationship to the field quality of the samples. In fact the samples taken at Scollard Canyon appeared to be perhaps the best of any in the whole sequence, yet provided the least consistency. One is tempted to believe that the differences are due to original magnetic characteristics of the sediment. Long periods of constant polarity may generally represent a stronger dipole field than during short-term polarity reversals. If so, polarity zone 30r is an exception. Polarity zone 29r gave the most scatter between samples. It might be noted here that the senior author encountered the same problem in this magnetozone in the cherty deep-sea limestone section on Woodside Creek, New Zealand. Perhaps the Earth's dipole field was relatively weak just at the end of the Cretaceous, but we are not aware of any similar problem being reported from other sections.

Biostratigraphy and geochronology

Campanian–Maastrichtian boundary

The Campanian–Maastrichtian foraminiferal boundary in the marine section at Gubbio, Italy, occurs about one fifth of the way down in the Gubbio B^+ zone, which is correlated to sea-floor anomaly 33 (Lowrie and Alvarez 1977*b*). The magnetostratigraphy of the Horseshoe Canyon Formation in the Red Deer Valley indicates that the basal part of the formation is low in polarity zone 33 and therefore should be Campanian (Fig. 17).

The biostratigraphic position of the Campanian–Maastrichtian boundary in North America has not been agreed upon between ammonite and foraminifera specialists (see Obradovich and Cobban 1975, Table 1). Jeletzky's (1968) correlation of western North American and European ammonite zones places the boundary at the base of the *Baculites baculus* Zone. Gill and Cobban (1973) accepted this placement of the boundary, but later Cobban (*in* Obradovich and Cobban 1975) suggested that it could be somewhat lower, perhaps as low as the base of the *B. reesidei* Zone. The fossil evidence in western Canada pertaining to this boundary is meager but supports Jeletzky's placement of the boundary at the base of the *B. baculus* Zone. Foraminiferal correlations between the South Saskatchewan River valley in Saskatchewan (350 km east of Drumheller, Alberta) and the Castor area of Alberta, 110 km north of Drumheller (Given and Wall 1971), show that the upper Bearpaw Shale is replaced westward from Saskatchewan by the deltaic lower Horseshoe Canyon Formation in central Alberta. The 25 m thick uppermost shale unit of the Bearpaw

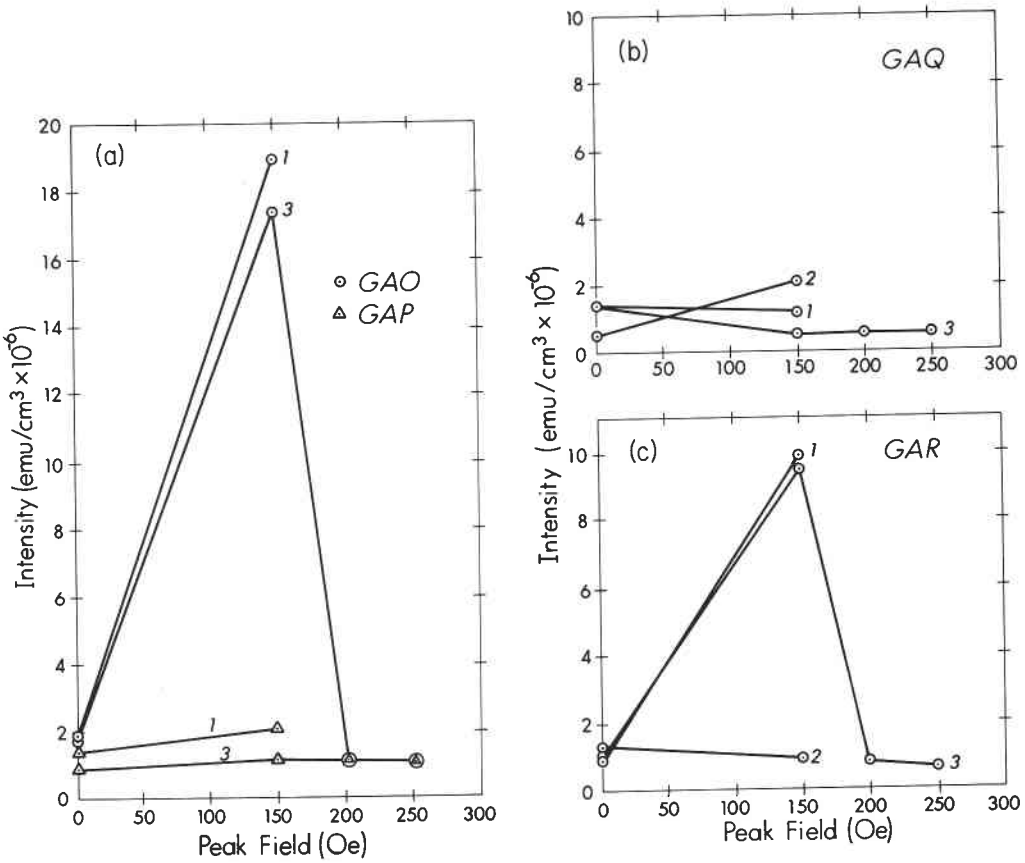


FIG. 8. Changes in intensity during alternating field demagnetization of a succession of four very fine sandstones GAO, GAP, GAQ, GAR, from polarity zone 30r in the east Scollard Canyon section. See Fig. 3 for stratigraphic position.

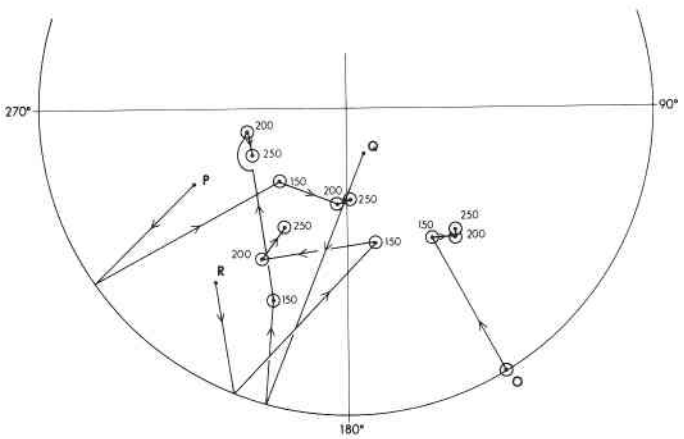


FIG. 9. Changes in direction during alternating field demagnetization of the third sample at each of four very fine sandstone sites, GAO, GAP, GAQ, GAR, in polarity zone 30r in the east Scollard Canyon section. See Fig. 3 for stratigraphic position.

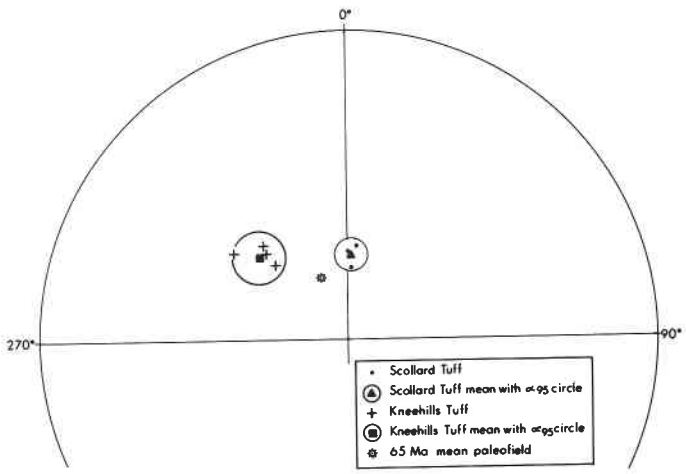


FIG. 10. Magnetic directions of four samples of the Kneehills Tuff and three samples of the Scollard tuff after AF demagnetization at 200 Oe (20 mT) peak field. Each sample is from an independently oriented hand block. The α_{95} circles of confidence around the means are 6.2 and 4.6°, respectively.

at Drumheller (Fig. 17) and Castor (Given and Wall 1971, Fig. 4) correlates with the *Baculites reesidei* Zone of the Bearpaw in the South Saskatchewan Valley (Caldwell 1968, Fig. 16; Given and Wall 1971, Fig. 4; Caldwell *et al.* 1978). As the magnetostratigraphy of the Horseshoe Canyon Formation indicates that the uppermost shale of the Bearpaw Formation is well down in polarity zone 33, the position of the Campanian–Maastrichtian boundary, which is well up in zone 33, must also be well above the *Baculites reesidei* Zone. An upper constraint on the boundary is provided by magnetostratigraphic correlation between the Drumheller area and the Eastend area of the Cypress Hills in southwestern Saskatchewan (Lerbekmo 1985). *Baculites grandis* has been found near Eastend (McLearn 1928). The present senior author has determined this location to be about 20 m below the top of the Bearpaw Shale. The top of the Bearpaw at this locality is in the lower part of polarity zone 31r, and consideration of relative thickness of the polarity zones indicates that the base of the *B. grandis* zone will

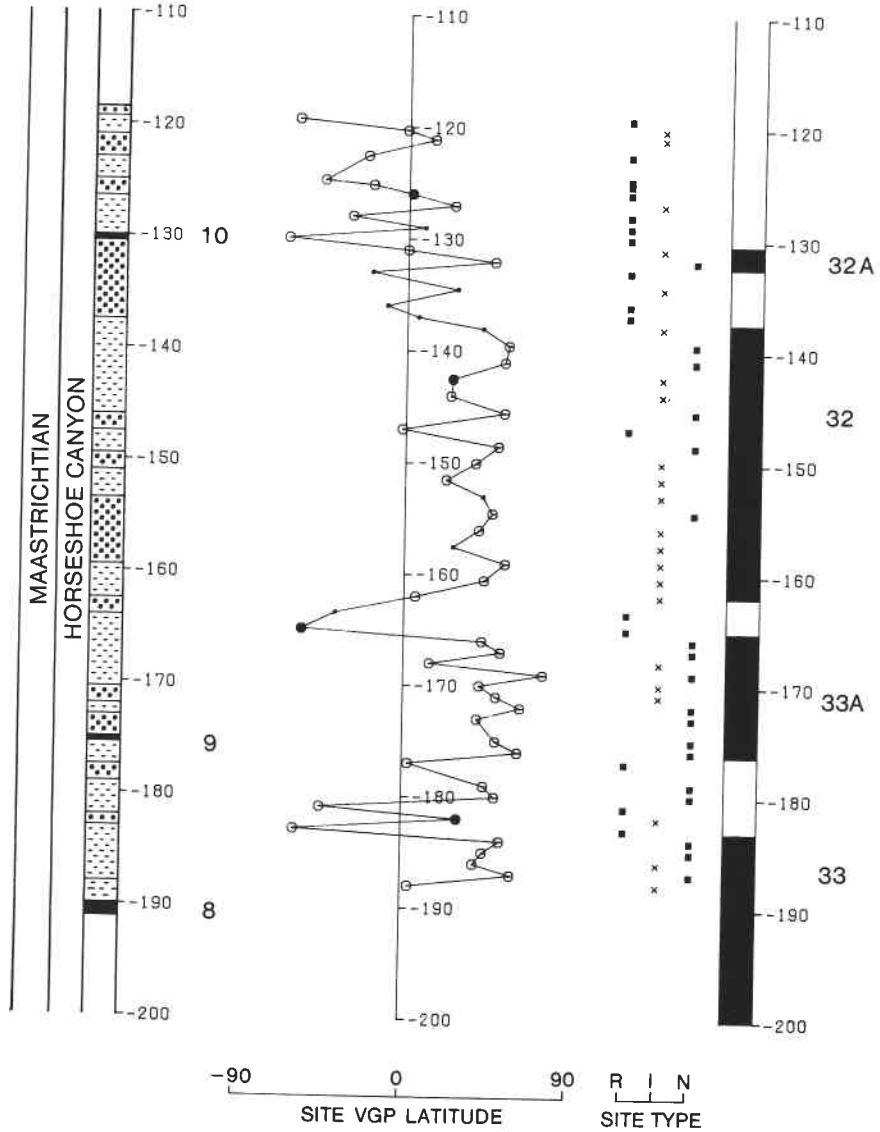


FIG. 11. Morrin Bone-bed section (NW 1/4, sec. 22, tp. 31, rge. 21, W 4th mer.). See Fig. 3 for legend.

be near the base of polarity zone 32 or in 32r. Therefore, the Campanian–Maastrichtian boundary, being high in polarity zone 33, is well above the *B. reesidei* Zone and well below the *B. grandis* Zone. As only two ammonite zones (*B. eliasi* and *B. baculus*) intervene (Obradovich and Cobban 1975), the base of the *B. baculus* Zone is the best choice magnetostratigraphically for the boundary. The base of polarity zone 32 is at −160 m in the Horseshoe Canyon Formation (Fig. 17). If the lowest 170 m of the formation is occupied by only the equivalent of the *B. baculus* and *B. eliasi* zones, the base of the *B. baculus* Zone would be expected to be equivalent to the level of the No. 5 or No. 6 coal seam. As the sequence is continental at this level, the biostratigraphic control is largely palynofloral. Srivastava (1970) recorded the lowest occurrence of *Wodehouseia*, namely, *W. jacutense* and *W. gracilis*, just below the No. 5 coal seam. These forms have been grouped under *Wodehouseia edmontonicola* by Wiggins (1976) who described this species also from the lower Maastrichtian of the Arctic North Slope, Alaska. As the genus *Wodehouseia* has only been reported from Maastrichtian and later strata (both in North America and Russia), its first appearance is taken to signal the crossing of the Campanian–Maastrichtian boundary (Wiggins 1976).

In summary, although the magnetostratigraphic and biostratigraphic correlations near the Campanian–Maastrichtian boundary in the Red Deer Valley are somewhat labored, they are mutually supportive and together strongly favor placing the boundary approximately at the level of the No. 5 coal seam, which is nearly halfway down in polarity zone 33 in this section.

The radiometric age of this boundary, if taken as the base of the *Baculites baculus* Zone, rests upon a number of K/Ar dates on sanidine and biotite from bentonites in the Bearpaw Shale of Alberta and Saskatchewan and the Pierre Shale of Wyoming (see Obradovich and Cobban 1975, Table 1). There is agreement on a date of 68.5 Ma for the base of the *Baculites grandis* Zone and 71.5 for the *B. compressus* Zone (old isotopic constants), a spread of 3 Ma encompassing six ammonite zones, or about 0.5 Ma/zone. This would suggest an age near 69 Ma for the base of the *B. baculus* Zone, or about 71 Ma when calculated with the new K/Ar constants (Steiger and Jäger 1977).

It will have been noted that there are some differences between these data and the interpretation presented by Harland *et al.* 1982 (Fig. 19).

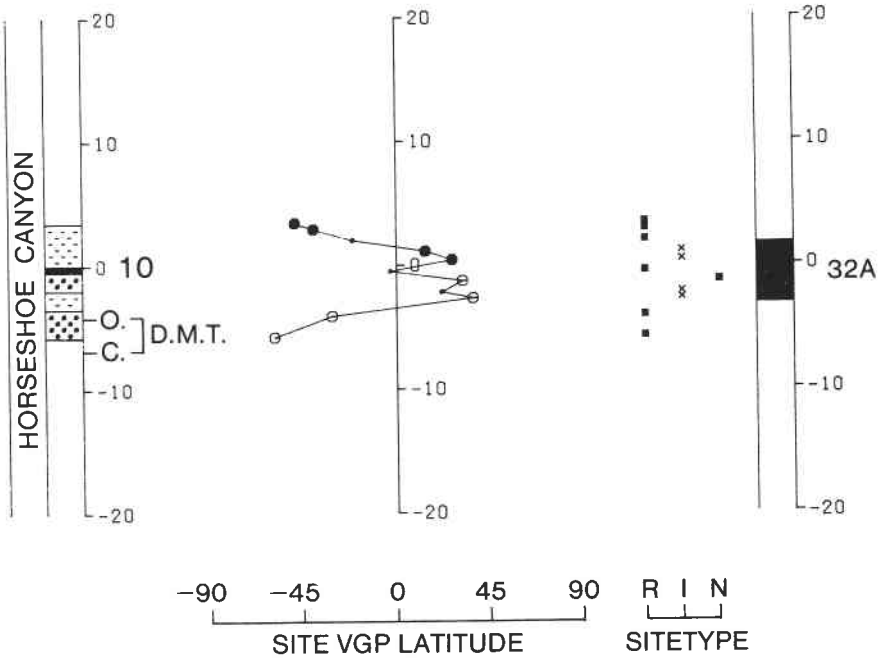


FIG. 12. Horsethief Canyon (auxilliary) section (SE 1/4, sec. 3, tp. 30, rge. 21, W 4th mer.). D.M.T. = Drumheller marine tongue; O. = *Ostrea*; C. = *Corbicula*. See Fig. 3 for additional legend.

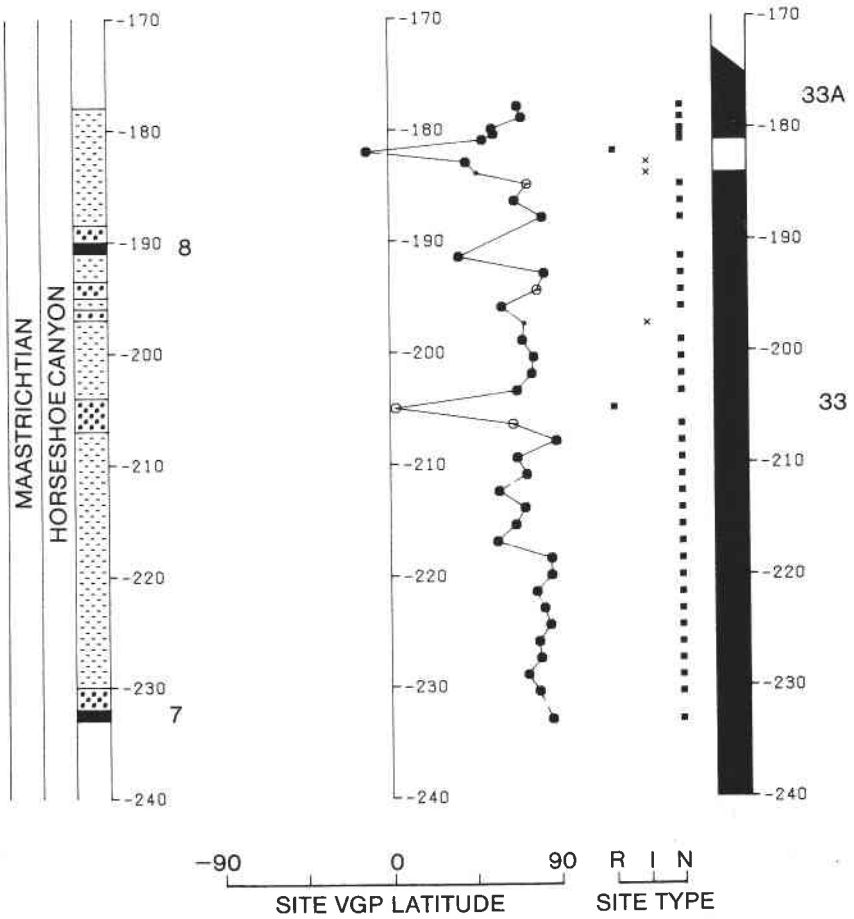


FIG. 13. Dinosaur Campground section (NW 1/4, sec. 25, tp. 29, rge. 21, W 4th mer.). See Fig. 3 for legend.

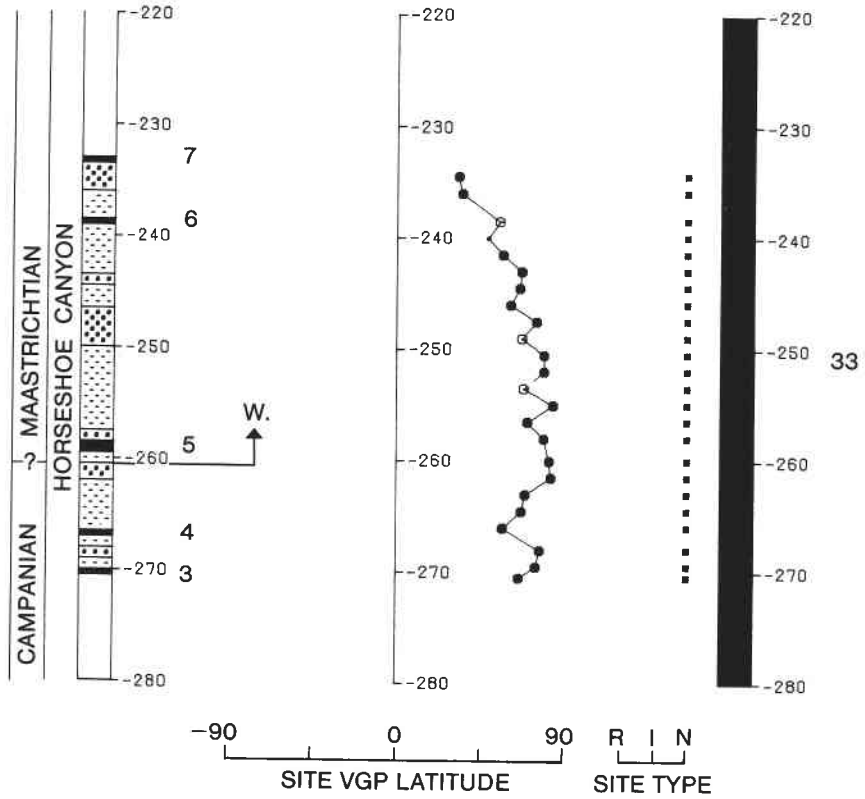


FIG. 14. *Gorgosaurus* Locality section (NE 1/4, sec. 36, tp. 28, rge. 20, W 4th mer.). W. = first appearance of *Wodehouseia*. See Fig. 3 for legend.

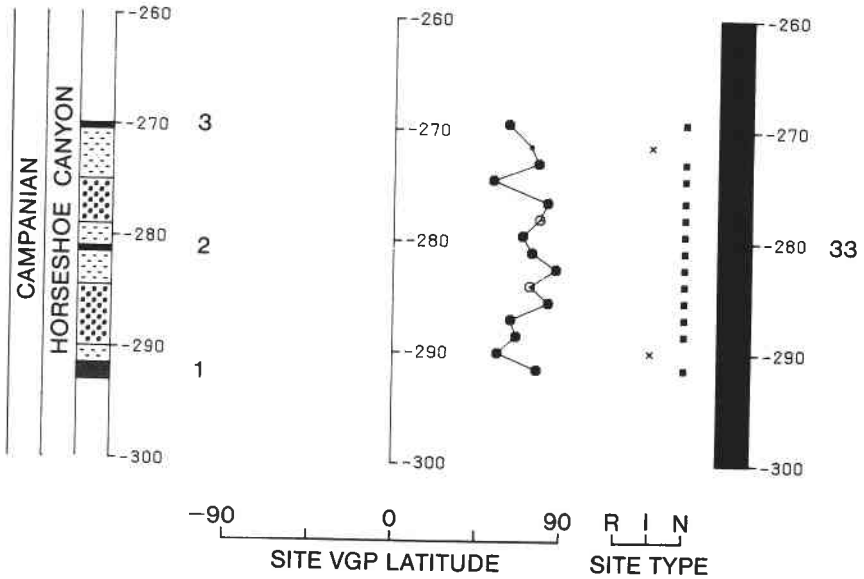


FIG. 15. Willow Creek section (NW 1/4, sec. 7, tp. 28, rge. 18, W 4th mer.). See Fig. 3 for legend.

Firstly, in the Harland *et al.* (1982) time scale the Campanian–Maastrichtian boundary occurs less than one tenth of the way down in polarity chron 33. However, the data of Lowrie and Alvarez (1977*b*, Fig. 9) and Alvarez *et al.* (1977, Fig. 2) from Gubbio place it no higher than one fifth of the way down in the polarity zone. Lowrie and Alvarez also showed the base of zone 33 with a range of uncertainty that could place the Campanian–Maastrichtian boundary proportionally even lower. The data from Alberta and Saskatchewan suggest that in the Red Deer Valley section the boundary is at least one third of the way down in zone 33.

Secondly, the radiometric age of this boundary is put at 73 Ma in the Harland *et al.* (1982) time scale. This number has been "adjusted" slightly from the Ness *et al.* (1980) value of 72.3 Ma to smooth out the fluctuations in ocean-floor spreading rates. The 72.3 Ma comes from a value of 70.5 Ma interpolated from Obradovich and Cobban's (1975, Table 1) dates near the Campanian–Maastrichtian boundary and updated with the new

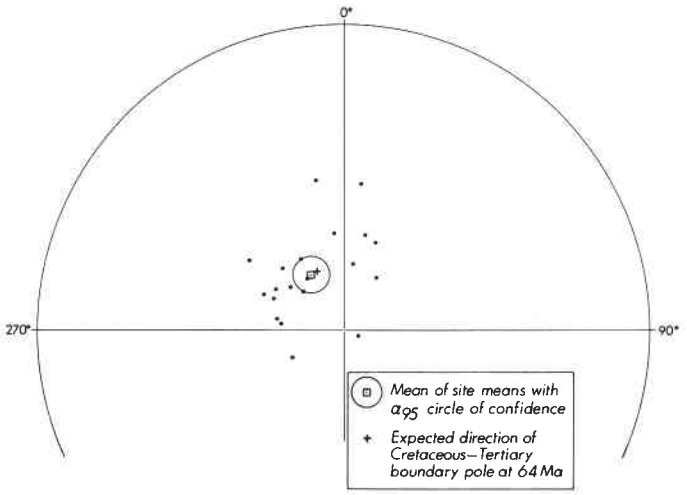


FIG. 16. Site mean directions and overall mean direction of 20 sites with $\alpha_{95} < 40°$ in the lowest 10 m of polarity zone 29 in the east Scollard Canyon section; α_{95} (mean of means) = 6.6°, k = 24.

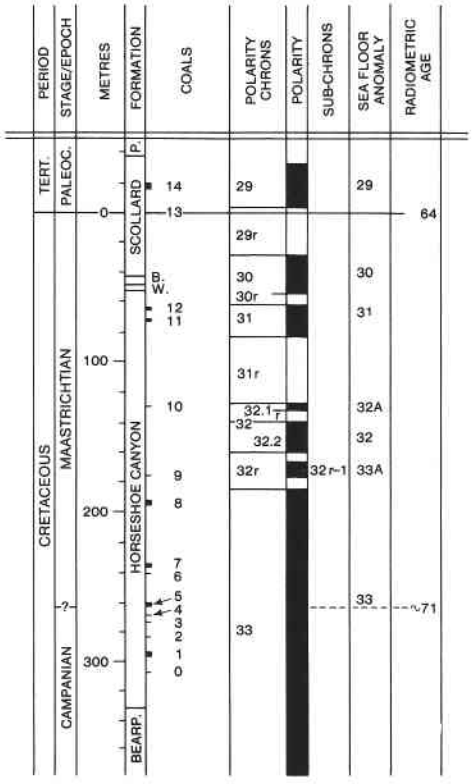


FIG. 17. Red Deer Valley composite magnetostratigraphic section. Polarity symbolism and nomenclature follow Harland *et al.* (1982).

constants. It represents a value for the base of the *Baculites reesidei* Zone rather than the *Baculites baculus* Zone.

Cretaceous–Tertiary boundary

Radiometric, dinosaur, and palynofloral data near the Cretaceous–Tertiary boundary in the Red Deer Valley have been given in earlier papers (Lerbekmo *et al.* 1979*a*,*b*). The boundary position was chosen at the highest occurrence of the palynomorph genus *Aquilapollenites*, which is restricted to the Cretaceous in Montana (Smit and van der Kaars 1984; Tschudy *et al.* 1984; Hotton 1984). In the Red Deer Valley, *Aquilapollenites* extends to 0.7 m above the top of the Nevis (No. 13)

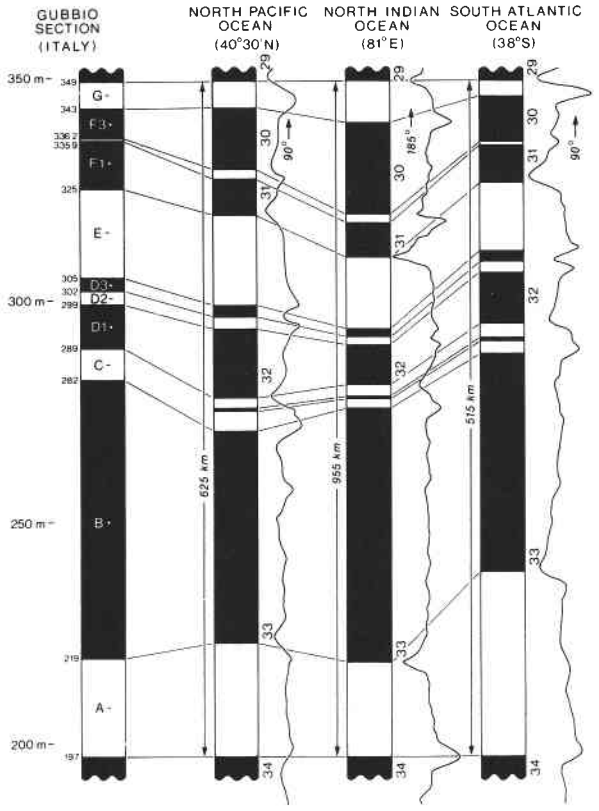


FIG. 18. Magnetostratigraphic correlations between sea floor magnetic anomalies and the section at Gubbio, Italy. (From Lowrie and Alvarez 1977*a*).

coal seam (C. Singh, personal communication, 1983). However, a search for the Cretaceous–Tertiary boundary iridium anomaly (Alvarez *et al.* 1980) was unsuccessful near this level. Recently, Sweet and Hills (1984, and personal communication, 1984) pointed out that the fern-spore peak associated with the boundary Ir anomaly in Montana (Smith and van der Kaars 1984) and Colorado – New Mexico (Orth *et al.* 1981; Tschudy *et al.* 1984) occurs in the lower part of the Nevis coal in the Red Deer Valley. Extending our search downwards, we have located the terminal Cretaceous Ir anomaly at the base of the Nevis coal (Lerbekmo and St. Louis, in preparation) and now place the Cretaceous–Tertiary boundary at this level (Fig. 3), which is approximately 4 m above the highest known dinosaur (*Triceratops*) occurrence in the area (Fig. 3*b*).

Radiometric dating of Campanian to Paleocene sediments in Alberta indicates a time span of about 7.5 Ma for the Maastrichtian, which in the Red Deer Valley is about 250 m thick. Thus the average rate of sedimentation is about 1 m/ 30 000 – 35 000 years, or 30 m/Ma. The age of the No. 14 (Ardley) coal seam is 63 Ma, based upon five sanidine-determined K/Ar ages from two bentonites (63.0, 63.0, 63.3; 62.8, 63.0 Ma) (Lerbekmo *et al.* 1979*b*). This coal seam is 18 m above the Cretaceous–Tertiary boundary, which would indicate an age of about 63.5 Ma for the boundary. This is in agreement with four K/Ar age determinations of 63.0, 63.3, 63.5, and 63.5 Ma on sanidines from the bentonite in the No. 13 (Nevis) coal (Lerbekmo *et al.* 1979*b*). The Ir anomaly at the base of the Z coal at the Hell Creek – Tullock formational boundary in Montana is about 3 m above the highest local occurrence of dinosaur bones (Baadsgaard and Lerbekmo 1980; Baadsgaard *et al.*, in preparation). A bentonite 0.7 m above the base of

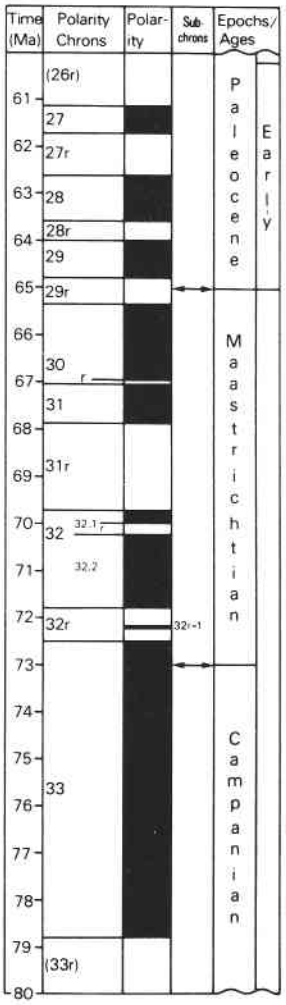


FIG. 19. Polarity time scale for late Campanian to Paleocene (from Harland *et al.* 1982).

this coal has given Rb/Sr ages on biotite of 63.7 ± 0.3 Ma and U/Pb ages on zircon of 63.9 ± 0.7 Ma (Baadsgaard and Lerbekmo 1980, 1983).

Conclusions

(1) The late Campanian to Early Paleocene fluvial–deltaic sediments of the Edmonton Group in the Red Deer Valley have recorded a geomagnetic polarity pattern very similar to that shown by sea-floor magnetic anomalies 29–33 and by the marine magnetostratigraphic "type section" at Gubbio, Italy.

(2) As a corollary to the first conclusion, fluvial–deltaic successions do not necessarily have any single hiatuses of stratigraphic consequence, and therefore for parts of the geologic column where polarity reversals are frequent magnetostratigraphy may be very useful as a general stratigraphic tool.

(3) All major fluvial–deltaic lithologies of medium sand grade and finer are probably suitable for initially recording detrital remanent magnetization and may be sampled systematically for routine work.

(4) The Campanian–Maastrichtian boundary, as determined by changes in palynoflora, and the Maastrichtian–Paleocene (Cretaceous–Tertiary) boundary, as marked by an Ir anomaly associated with pronounced extinctions of palynoflora and reptiles, are closely coincident with these stage boundaries in marine sections where they are determined by ammonite and foraminiferal extinctions. Radiometric data to date from southern Canada and the northern United States indicate the ages of these boundaries to be approximately 71 and 64 Ma, respectively.

Acknowledgments

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